



Problem 1 Fourier Series (10 points) Consider the periodic function $f(t)$ with period 2π , defined by:

$$f(t) = \begin{cases} 1, & 0 < t < \pi, \\ -1, & -\pi < t < 0, \\ 0 & t = 0, \pm\pi, \end{cases}$$

and extended periodically outside $-\pi < t < \pi$.

- Determine whether $f(t)$ is even, odd, or neither. Explain your reasoning and what this means for the Fourier series of $f(t)$.
- Compute the Fourier series of $f(t)$, giving explicit expressions for the Fourier coefficients a_0, a_n, b_n .
- Write the first three nonzero terms of the Fourier series explicitly.

Solution:

a) **Step 1: Check parity on the base interval $(-\pi, \pi)$**

- For $t \in (0, \pi)$:

$$f(-t) = f(\text{value in } (-\pi, 0)) = -1 = -f(t).$$

- For $t \in (-\pi, 0)$:

$$f(-t) = f(\text{value in } (0, \pi)) = 1 = -f(t).$$

- At $t = 0$, $f(0) = 0$ and $-f(0) = 0$, so the parity relation holds.

Conclusion: f is odd on $[-\pi, \pi]$.

Step 2: Extend the parity check to all $t \in \mathbb{R}$

- f is periodic with period 2π . - For a periodic function, if it is odd on one full period, it remains odd for all $t \in \mathbb{R}$.

Conclusion: f is odd for all real numbers.

Step 3: Consequence for the Fourier series

- Since f is odd, all cosine coefficients vanish:

$$a_0 = 0, \quad a_n = 0 \quad \text{for all } n \geq 1.$$

- Only sine coefficients b_n may be nonzero. - Therefore, the Fourier series of f contains **only sine terms**.

b) For a 2π -periodic function:

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt$$

Since $f(t)$ is odd, we can integrate from 0 to π and double the integral:

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(t) \sin(nt) dt$$

On $0 < t < \pi$, $f(t) = 1$:

$$b_n = \frac{2}{\pi} \int_0^{\pi} \sin(nt) dt = \frac{2}{\pi} \left[-\frac{\cos(nt)}{n} \right]_0^{\pi}$$

$$b_n = \frac{2}{\pi} \left(-\frac{(-1)^n - 1}{n} \right) = \frac{2}{\pi} \cdot \frac{1 - (-1)^n}{n}$$

- For even n : $b_n = 0$

- For odd n : $b_n = \frac{4}{n\pi}$.

c) The Fourier series is:

$$f(t) = \sum_{n=1,3,5,\dots}^{\infty} \frac{4}{n\pi} \sin(nt)$$

First three nonzero terms ($n = 1, 3, 5$):

$$f(t) \approx \frac{4}{\pi} \sin(t) + \frac{4}{3\pi} \sin(3t) + \frac{4}{5\pi} \sin(5t)$$

Grading manual: a) 3 points (1 points for f odd in $(-\pi, \pi)$, 1 points for the rest of $\mathbb{R}$, 1 point for the consequence for the Fourier coefficients) b) 5 points c) 2 points

Problem 2 Interpolation (10 points)

The following table shows the values of a function $f(x)$ at the following points:

x	$f(x)$
1	2
2	8
3	18
4	32

- Construct an interpolating polynomial $p(x)$ of minimal necessary degree for the given data using the **Lagrange interpolation formula**. Simplify the resulting polynomial as much as possible.
- Use $p(x)$ to compute an approximate value of $f(2.5)$.
- Briefly explain why the interpolating polynomial of minimal necessary degree exists and is unique for this data set.

Solution:

a) We are given data points (x_i, y_i) for $i = 1, 2, 3, 4$:

$$(1, 2), \quad (2, 8), \quad (3, 18), \quad (4, 32).$$

The Lagrange interpolant of degree at most 3 is

$$P(x) = \sum_{i=1}^4 y_i \ell_i(x), \quad \ell_i(x) = \prod_{\substack{j=1 \\ j \neq i}}^4 \frac{x - x_j}{x_i - x_j}.$$

With nodes $x_1 = 1$, $x_2 = 2$, $x_3 = 3$, $x_4 = 4$, the basis polynomials are

$$\begin{aligned} \ell_1(x) &= \frac{(x-2)(x-3)(x-4)}{(1-2)(1-3)(1-4)} = \frac{(x-2)(x-3)(x-4)}{-6}, \\ \ell_2(x) &= \frac{(x-1)(x-3)(x-4)}{(2-1)(2-3)(2-4)} = \frac{(x-1)(x-3)(x-4)}{2}, \\ \ell_3(x) &= \frac{(x-1)(x-2)(x-4)}{(3-1)(3-2)(3-4)} = \frac{(x-1)(x-2)(x-4)}{-2}, \\ \ell_4(x) &= \frac{(x-1)(x-2)(x-3)}{(4-1)(4-2)(4-3)} = \frac{(x-1)(x-2)(x-3)}{6}. \end{aligned}$$

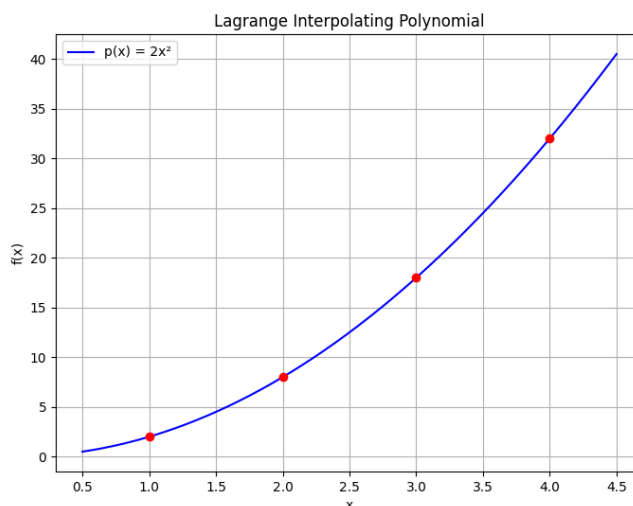
Thus

$$p(x) = 2\ell_1(x) + 8\ell_2(x) + 18\ell_3(x) + 32\ell_4(x).$$

Expanding and simplifying gives

$$p(x) = 2x^2.$$

(Indeed, the data satisfy $y = 2x^2$ at $x = 1, 2, 3, 4$.)



b) **Estimate at $x = 2.5$:**

$$f(2.5) \approx P(2.5) = 2(2.5)^2 = 2 \cdot 6.25 = 12.5.$$

c) **Existence and Uniqueness:** For **distinct** nodes $x_1, \dots, x_4$, there exists a unique polynomial of degree ≤ 3 that interpolates the data according to the lectures. The uniqueness theorem applies whenever the interpolation nodes are distinct. If two nodes coincide with different function values, no interpolating polynomial exists. (This is enough for two points.)

Note: If q were another such polynomial, then $r = p - q$ would be a nonzero polynomial of degree ≤ 3 with four distinct roots $x_1, \dots, x_4$, which is impossible. Hence p is unique.

Grading manual: a) 7 points b) 1 point c) 2 points

Problem 3 Numerical Integration (10 points)

a) Consider the approximation formula

$$\int_0^h f(s)ds \approx h \left(A_1 f(0) + A_2 f(h/3) + A_3 f(h) \right).$$

Find coefficients $\{A_i\}_{i=1}^3$ such that the formula becomes exact for all polynomials in P_2 , i.e., polynomials of degree less than or equal to 2.

b) i) The Simpson's rule provides an approximation for

$$\int_a^b f(s)ds = \frac{\tilde{h}}{3} \left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right) - \frac{\tilde{h}^5}{90} f^{(4)}(\tilde{\zeta})$$

where $2\tilde{h} := (b - a)$ and $\tilde{\zeta}$ lies between a and b .

Starting from an equally spaced partition of the interval $[a, b]$

$$a = x_0 < x_1 < x_2 < \cdots < x_{2m-2} < x_{2m-1} < x_{2m} = b,$$

the composite Simpson's rule is defined by applying the Simpson's rule to each subinterval $[x_{2i}, x_{2i+2}]$, and the results are combined to approximate an integral over the whole interval.

Show that the composite Simpson's rule has the following form

$$\int_a^b f(s)ds = \frac{h}{3} \left(f(a) + f(b) + 4 \sum_{i=0}^{m-1} f(x_{2i+1}) + 2 \sum_{i=1}^{m-1} f(x_{2i}) \right) - m \frac{h^5}{90} f^{(4)}(\zeta).$$

where $2h := (x_{2i+2} - x_{2i}) = \frac{b-a}{m}$ and ζ lies between a and b .

ii) Use the composite Simpson's rule for $2m = 8$ to compute an approximation for

$$\int_1^2 \ln(s)ds = 2 \ln(2) - 1.$$

iii) Show that the approximation error cannot be larger than 0.000008.

Solution

- a) Linearity implies that, to establish exactness of the formula for all polynomials of degree at most 2, it suffices to verify it for the basis polynomials $\{1, x, x^2\}$. This results in the set of equations

$$1 = A_1 + A_2 + A_3 \quad (1)$$

$$1/2 = A_2/3 + A_3 \quad (2)$$

$$1/3 = A_2/9 + A_3 \quad (3)$$

Solving this system gives $A_1 = 0$, $A_2 = 3/4$, $A_3 = 1/4$.

- b) i) We consider an equally spaced partition of the interval $[a, b]$

$$a = x_0 < x_1 < x_2 < \cdots < x_{2m-2} < x_{2m-1} < x_{2m} = b$$

We apply the Simpson's rule on each interval $[x_{2i}, x_{2i+2}]$, $i = 0, \dots, m$

$$\int_a^b f(s)ds = \sum_{i=0}^{m-1} \int_{x_{2i}}^{x_{2i+2}} f(s)ds \quad (4)$$

$$= \sum_{i=0}^{m-1} \left(\frac{h}{3} \left(f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2}) \right) - \frac{h^5}{90} f^{(4)}(\zeta_i) \right) \quad (5)$$

$$= \frac{h}{3} \left(f(a) + f(b) + 4 \sum_{i=0}^{m-1} f(x_{2i+1}) + 2 \sum_{i=1}^{m-1} f(x_{2i}) \right) - \sum_{i=0}^{m-1} \frac{h^5}{90} f^{(4)}(\zeta_i) \quad (6)$$

where $2h = x_{2i} - x_{2i+2}$ and ζ_i lies between x_{2i} and x_{2i+2} . Using the generalized intermediate value theorem (used in the lecture)

$$\sum_{i=0}^{m-1} \frac{h^5}{90} f^{(4)}(\zeta_i) = m \frac{h^5}{90} f^{(4)}(\zeta),$$

where $a < \zeta < b$.

- ii) Using the composite Simpson's rule for $2m = 8$, we find

$$\int_1^2 \ln(s)ds \approx \frac{(2-1)/8}{3} \left(\ln(1) + \ln(2) + 4 \ln(9/8) + 4 \ln(11/8) + 4 \ln(13/8) \right. \quad (7)$$

$$\left. + 4 \ln(15/8) + 2 \ln(5/4) + 2 \ln(6/4) + 2 \ln(7/4) \right) \approx 0.386292. \quad (8)$$

- iii) We observe that this matches within five decimal places with the value 0.386294 computed in ii). The approximation error is bounded by

$$\frac{h^4}{180} |f^{(4)}(\zeta)| = \frac{(1/8)^4}{180} \frac{6}{\zeta^4} \leq \frac{6}{8^4 \cdot 180} \approx 0.000008.$$

Grading manual: a) 4 points b)i) 3 points b)ii) 1 point b)iii) 2 points

Problem 4 Fourier Transform (10 points)

a) Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(t) = \begin{cases} -1 & -1 < t < 0 \\ 1 & 0 < t < 1 \\ 0 & \text{else} \end{cases}$$

Compute its Fourier transform

$$\mathcal{F}[f](w) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(s) \exp(-iws) ds.$$

Note that $\exp(-iws) = e^{-iws}$.

b) From the lectures you know that

$$\mathcal{F}[g'](w) = iw\mathcal{F}[g](w),$$

where g' denotes the derivative of g . Compute the Fourier transform of the function

$$f(t) = \frac{t}{(t^2 + c^2)^2}, \quad c > 0.$$

Hint: Use the fact that the Fourier transform of the function $h(t) = c(t^2 + c^2)^{-1}$ is the function $\hat{h}(w) = \sqrt{\pi/2} \exp(-c|w|) = \sqrt{\pi/2} e^{-c|w|}$.

Solution

a) First note that f is an odd function between -1 and 1 . Thus, the product $f(t) \cos(wt)$ is an odd function, whereas the product $f(t) \sin(wt)$ is an even function.

Next, recall that

$$\exp(-iws) = \cos(ws) - i \sin(ws).$$

The computation of the Fourier transform then yields

$$\mathcal{F}[f](w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(s) \exp(-iws) ds \quad (9)$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 f(s) \exp(-iws) ds \quad (10)$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 f(s) \cos(ws) ds - \frac{i}{\sqrt{2\pi}} \int_{-1}^1 f(s) \sin(ws) ds \quad (11)$$

$$= -\frac{i}{\sqrt{2\pi}} \int_{-1}^1 f(s) \sin(ws) ds \quad (12)$$

$$= -\frac{2i}{\sqrt{2\pi}} \int_0^1 \sin(ws) ds \quad (13)$$

$$= \frac{2i}{\sqrt{2\pi}} \frac{\cos(ws)}{w} \Big|_0^1 \quad (14)$$

$$= i \sqrt{\frac{2}{\pi}} \frac{\cos(w) - 1}{w} \quad (15)$$

Observe that this formula holds for w different from zero. The case $w = 0$ can be computed separately, giving $\mathcal{F}[f](0) = 0$.

b) Define the function

$$g(t) = \frac{1}{t^2 + c^2}$$

and observe that

$$f(t) = -\frac{1}{2} \frac{d}{dt} g(t).$$

The result from the lecture together with the hint then imply that

$$\mathcal{F}[g'](w) = iw \mathcal{F}[g](w) \quad (16)$$

$$= iw \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{s^2 + c^2} \exp(-iws) ds \quad (17)$$

$$= -\frac{iw}{2c} \sqrt{\frac{\pi}{2}} \exp(-c|w|). \quad (18)$$

Grading manual: a) 4 points and 1 point for $w = 0$ case b) 5 points

Problem 5 Laplace Transform (10 points)

- a) Compute the Laplace transform of

$$f(t) = t \cos(2t).$$

- b) Compute the inverse Laplace transform of

$$F(s) = \frac{3s + 5}{(s - 3)(s + 1)}$$

- c) Use the Laplace transform to solve the initial value problem

$$y'(t) + 3y(t) = \sin(2t), \quad y(0) = 1.$$

Solution

- a) Use $\mathcal{L}\{tf(t)\} = -\frac{d}{ds}\{\mathcal{L}\{f(t)\}\}$ and $\mathcal{L}\{\cos(2t)\}(s) = \frac{s}{s^2 + 4}$. Then

$$\mathcal{L}\{t \cos(2t)\}(s) = -\frac{d}{ds} \left(\frac{s}{s^2 + 4} \right).$$

Differentiate:

$$\frac{d}{ds} \left(\frac{s}{s^2 + 4} \right) = \frac{(s^2 + 4) - 2s^2}{(s^2 + 4)^2} = \frac{-s^2 + 4}{(s^2 + 4)^2}.$$

So

$$\mathcal{L}\{t \cos(2t)\}(s) = \frac{s^2 - 4}{(s^2 + 4)^2}.$$

- b) We use a partial fraction decomposition

$$\frac{3s + 5}{(s - 3)(s + 1)} = \frac{A}{s - 3} + \frac{B}{s + 1},$$

so

$$3s + 5 = A(s + 1) + B(s - 3),$$

which gives $A = \frac{7}{2}$ and $B = -\frac{1}{2}$. Thus

$$\mathcal{L}^{-1}\{F(s)\} = \frac{7}{2}e^{3t} - \frac{1}{2}e^{-t}.$$

c) Taking the Laplace transform of the initial value problem gives

$$(sY(s) - 1) + 3Y(s) = \frac{2}{s^2 + 4}.$$

So

$$(s + 3)Y(s) - 1 = \frac{2}{s^2 + 4}, \quad Y(s) = \frac{1}{s + 3} + \frac{2}{(s + 3)(s^2 + 4)}.$$

Using partial fraction composition on the second term gives

$$\frac{2}{(s + 3)(s^2 + 4)} = \frac{A}{s + 3} + \frac{Bs + C}{s^2 + 4}.$$

with solution $A = \frac{2}{13}$, $B = -\frac{2}{13}$, $C = \frac{6}{13}$. So

$$Y(s) = \frac{1}{s + 3} + \frac{2}{13} \frac{1}{s + 3} - \frac{2}{13} \frac{s}{s^2 + 4} + \frac{6}{13} \frac{1}{s^2 + 4}.$$

By table we can find the inverse transform, that is, the solution

$$y(t) = \frac{15}{13}e^{-3t} - \frac{2}{13}\cos(2t) + \frac{3}{13}\sin(2t).$$

Grading manual

a) (2p)

b) (4p)

c) (4p)

Problem 6 Heat Equation (10 points)

Consider the heat equation

$$u_t(x, t) = 6u_{xx}(x, t), \quad 0 < x < \pi, \quad t > 0,$$

with boundary conditions

$$u(0, t) = u(\pi, t) = 0,$$

and initial condition

$$u(x, 0) = 3 \sin x + 2 \sin(2x) + \sin(3x).$$

- a) Use the separation of variables to find the non-trivial solutions $u_n(x, t)$ to the heat equation with the given boundary conditions.
- b) Use the initial condition to determine the full solution.
- c) What is the behavior of $u(x, t)$ when $t \rightarrow \infty$. Explain briefly how this relate to the physical interpretation of the boundary conditions if we assume that these equations model a metal rod between $x = 0$ and $x = \pi$ with temperature u ? What is the behavior of $u(x, t)$ when $t \rightarrow \infty$. Explain briefly how this relate to the physical interpretation of the boundary conditions if we assume that these equations model a metal rod between $x = 0$ and $x = \pi$ with temperature u ?

Solution

- a) Assume $u(x, t) = X(x)T(t)$. Then

$$\frac{T'}{6T} = \frac{X''}{X} = -\lambda.$$

So $X'' + \lambda X = 0$ with $X(0) = X(\pi) = 0$. We only find nontrivial solutions when $\lambda = n^2$ and these are $X_n(x) = \sin(nx)$. This gives the equation

$$T' = -6n^2T,$$

which has general solution $T_n(t) = e^{-6n^2t}$. This gives the eigenfunctions

$$u_n(x, t) = X_n(x)T_n(t) = \sin(nx)e^{-6n^2t}.$$

b) The solution is a series of eigenfunctions

$$u(x, t) = \sum_{n=1}^{\infty} b_n u_n(x, t) = \sum_{n=1}^{\infty} b_n \sin(nx) e^{-6n^2 t}.$$

From the initial condition we see that $b_1 = 3$, $b_2 = 2$, $b_3 = 1$. So the general solution is

$$u(x, t) = 3e^{-6t} \sin x + 2e^{-24t} \sin(2x) + e^{-54t} \sin(3x).$$

c) We find that

$$\lim_{t \rightarrow 0} u(x, t) = 0,$$

which is consistent with the interpretation of the boundary conditions that the rod is in contact with some heat conducting material with temperature held fixed at 0 at the endpoints $x = 0$ and $x = \pi$. In such a physical system the rod will eventually reach thermodynamic equilibrium with the exterior and attain the temperature 0 uniformly.

Grading manual

a) (5p)

b) (3p)

c) (2p)

Grading Scale (The following grading scheme is adjusted from this one:
<https://i.ntnu.no/wiki/-/wiki/English/Grading+scale+using+percentage+points>):

A	54–60
B	47–53
C	40–46
D	33–39
E	25–32
F	0–24